



DEALER BEST PRACTICE SERIES

Measuring Empty Machine Weight (EMW)

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

**MARC
Management**

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1.0 Introduction

Off Highway truck payload management – “10/10/20” is a fundamental requirement for proper equipment management and productivity management. To comply with the Truck Payload Policy, the customer, dealer and Caterpillar will need to know the empty weight of the truck. Due to the many optional attachments offered to the customer, there might be a discrepancy between the actual delivered Empty Machine Weight (EMW) and the EMW supplied by the factory. In some instance, it can be as high as 5 - 7% of target payload. Such discrepancies may lead to potential problem when the EMW is measured via production study scale down the track.

The Best Practice is to introduce Empty Machine Weight (EMW) measurement as part of the new machine delivery process. It will eliminate the potential ramification, allow us to meet our customer's expectation and allow the customer to accurately comply with the “10/10/20” payload policy.

2.0 Best Practice Description

It is important for the dealer sales representative to use the tools available to estimated EMW of the truck prior to taking an order from the customer. The best tool for this is the Mass Configurator and you can obtain this from your Caterpillar mining sales representative. This document will establish the target payload as defined by the stated configuration such that any changes in configuration during or after the sales process can be taken into account and the target payload can be adjusted accordingly. There are a few tools available for dealers to measure EMW.

Caterpillar Jack Stand

In some countries, Caterpillar Global Mining has a set of jack stands (see figure 2.1) for dealer to use, charges vary by region. Please check with your local Cat product support representative for further details and asset availability.



Figure 2.1: Caterpillar jack stand with load cell to measure machine empty weight

The jack stands are Caterpillar hydraulic lift with load cell for empty weight measurement. A trained and certified operator should operate the asset. Training and certification is available via your local Caterpillar Representative. Alternatively, Caterpillar will be able to run the empty weight study at a cost.

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Transcale Tare Weight Certification System

There are other empty weight measurement systems available in the market. For instance, Transcale Pty Ltd, Australia, can supply an empty weight certification system (See figure 2.2).

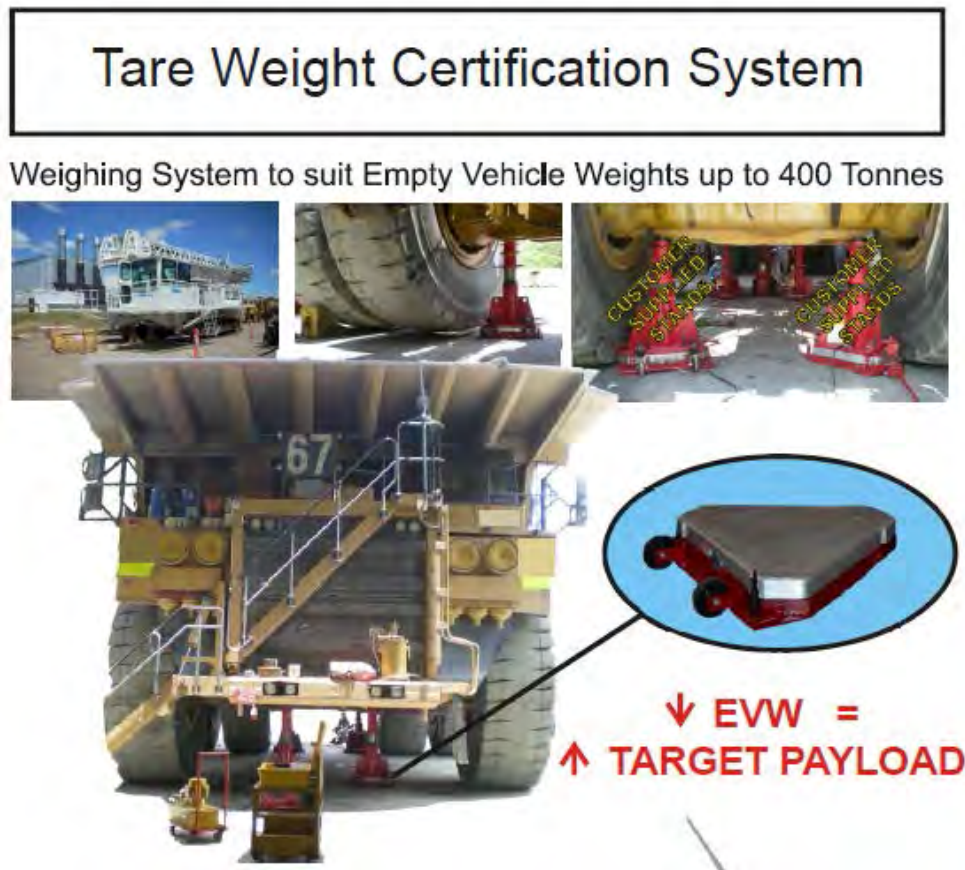


Figure 2.2: Transcale Empty Weight Certification System

The Transcale “Tare Weight Certification System” (TWCS) is basically a triangular shaped load cell unit that can be installed under the hydraulic lift device as showed in figure 2.2.

It is dealer’s discretion to purchase the TWCS and the cost can be justified by the elimination of potential ramifications down the track when the EMW is measured via a production study scale. Ideally, every truck delivered to the customer should have EMW measured.

Regardless of the tool chosen, it is important to formally document the empty weight. A certificate can be issued to the customer for each truck measured. Please refer to the attached document for certificate template and associate documents.

Note **Target Payload is the difference between Gross Machine Weight (GMW) and EMW. EMW includes body, body liner, attachments, tires and etc that are installed on the machines.**

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3.0 Implementation Steps

1. Organize the jack stands or TWCS
2. Organize a certificated operator for the study
3. Park the truck on a flat and level concrete floor suited for the weighing procedure
4. Capture optional attachment and relevant truck information (see attached documents)
5. Measure machine empty weight
6. Establish target payload based on EMW
7. Setup target payload and overload limits in VIMS
8. Generate and issue certificate for each truck being measured to the customer. The certificate should include the following information (see attached documents):
 - a. Truck body sales diagram detailing body and liner specification with associated weight estimate
 - b. Tires information including manufacturer, size, model and % tread depth
 - c. Percentage of fuel in the fuel tank
 - d. VIMS target payload based on the measured empty truck weight
9. Hand over the machine with the certificate
10. Send the copies of the certificates to Caterpillar
11. Caterpillar will verify measured machine weight with the “Mass Configuration Calculator”
 - a. This would allow Caterpillar to refine the mass configuration calculator and hence allowing us to provide more accurate EMW estimate to dealers
12. Keep record of all certificates to prevent potential problem down the track

4.0 Benefits

There have been multiple instances in the past where a fleet that has been in use for some time and has been found via a scale study to weigh more than expected. As the EMW becomes known, Caterpillar and the dealer are obliged to inform the customer of their true target payload, which is often less than their expectations. This can often degrade into an argument about how much the truck weighed at commissioning versus how much weight has since been added by the customer. Implementation of this Best Practice will nullify this argument. Typical weight additions include additional body liners, body reinforcements, debris/silt inside body rails, tire changes, rim changes, larger fuel tanks, etc. All of which are often not accounted for by the customer over the life of the truck. And if the VIMS target payload is not adjusted to account for the additional EMW then payload history indicating 10/10/20 compliance may in fact constitute significant overloading of the truck including possible breach of steering and braking certification limits.

It is also critically important to ensure machine EMW is kept under the certified ROPS limit. Without an EMW certificate at time of delivery the “growth” in EMW of a truck over time is likely to degrade into an argument with the customer over whether the truck was overweight at the time of commissioning.

5.0 Resources Required

- ☐ A Trained Operator
- ☐ Jack stands or TWCS
- ☐ Empty weight certificate template

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6.0 Supporting Attachments / References

- Generic_797F MSDII Body 252 m3.pdf
- Tare Weight Certificate Generic.doc

7.0 Related Best Practices

- 0707-1.1-1089 Payload Management

8.0 Acknowledgements

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